

Vikram Raj Nagoor Kani

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EDUCATION

University of Illinois Urbana-Champaign

Master of Engineering in Autonomy and Robotics

Urbana, IL

Aug 2024 - present

BS Abdur Rahman Crescent Institute of Technology

Bachelor of Technology in Electronics and Communication Engineering

Chennai, India

Jul 2017 - Jun 2021

TECHNICAL SKILLS

Programming Languages : Python, C++, Bash, PowerShell, LabView, Matlab, Azure CLI.

Technologies/Frameworks : Pytorch, Open CV, ROS, ROS2, Docker, Anaconda , NVIDIA Jetson, GEM E2, GEM E4, F1Tenth, Azure, Gazebo, Gazebo Ignition, Unity, Git, Linux, YOLO, Detic, SigLIP, Open AI Gym, Deep Learning, Machine Learning, UR3e.

Coursework : Principles of Safe Autonomy, Deep Learning with Computer Vision, Autonomous Vehicle System Engineering.

WORK EXPERIENCE

UIUC Research Lab, Robotics Intern at McKechnie Family LIFE Home

Jan 2025 - present

- Developing interactive demos for researchers to showcase **STRETCH robot** capabilities in assisting the elderly with household tasks in an **IoT-enabled** home using a combination of **LLMs and deep learning models**.
- Designing strategies for **distributed autonomous system coordination** to optimize task efficiency with STRETCH.

Accenture, Azure DevOps Automation Engineer

Aug 2021 - Jul 2024

- Developed and led **automation development** in the project, implementing Azure Runbooks for password rotation, vulnerability fixes and maintenance for cloud infrastructure and BareMetal physical servers. **Reducing manual hours spent by 90%**.
- Led deployment and testing of **cognitive vision models** on **IoT edge servers** at Microsoft Datacenters globally along side of Azure Cognitive services development team.

Nokia, Embedded System Engineer, Intern

Feb 2021 - May 2021

- Developed an **IoT device** to detect obstacles blocking accessibility of fire extinguishers and alert security for Nokia's Manufacturing Factory as part of their safety measures.

PROJECTS

LiDAR + Camera Based Navigation for Unpredictable Lighting Scenarios 🔗

ROS, Python, Open CV, NVIDIA Jetson, F1 Tenth, Unity, Linux

- Designed and implemented a lane detection algorithm using **OpenCV** on **Nvidia Jetson**, incorporating techniques such as down-scaling, resizing, color thresholding, and pixel intensity histograms and achieved an **30% increase in computational speed** compared to traditional methods, enabling **real-time waypoint generation**.
- Developed a robust switching mechanism between **camera-based error generation** and **LIDAR-based error generation** to handle low-light conditions, such as tunnels.
- Customized an open-source simulator using Unity to develop a **digital twin** of the lab environment and the F1TENTH car.

Persistent Pedestrian Detection Using Sensor Fusion 🔗

GEM E4, PyTorch, ROS, Python, Open CV, YOLO, GEMStack

- Developed a **pedestrian detection and tracking algorithm** utilizing camera and LiDAR data from a GEM e4 vehicle for **real-time autonomous systems**. Optimized computational efficiency by implementing **voxel down-sampling** to improve time complexity.
- Estimated **pedestrian velocity** by extracting pixel positions from camera data and applying the **RANSAC algorithm** to LiDAR points. Transformed detected pedestrian positions from the camera frame to the vehicle base frame and subsequently to the world frame, enabling accurate velocity calculation.

Contour-Based Image Drawing with UR3 Arm 🔗

ROS, Python, Open CV, UR3 Arm

- Designed an algorithm with **OpenCV** to detect **contours** from digital images, fine-tuned to **retain key features** and reduce waypoints for **faster drawing speed**. **Reducing time taken by 83%**.
- Converted pixel-based waypoints into real-world coordinates and applied **inverse kinematics** to accurately replicate the image with the UR3 arm.

Customized Object Detection using pre trained Neural Network models 🔗

PyTorch, Python, Open CV, CNN

- Customized a pre-trained ResNet50 Convolutional Neural Network model to detect breeds of cats and dogs. The dataset used was the Oxford-IIT Pet dataset, which was modified by **adding 7 more dog breeds**.
- The model was customized using the fast.ai deep learning library and **achieved 91% accuracy**.